

Auteur: Luko De Vriendt
Studierichting: Informatica
Oefeningenreeks 5E nr. 1

$$y(t) = y(-t)$$

$$\Leftrightarrow 4t - t^3 = -4t + t^3$$

$$\Leftrightarrow 8t - 2t^3 = 0$$

$$\Leftrightarrow 2t(4 - t^2) = 0$$

$$t = 0 \text{ of } t = \pm 2$$

$$\begin{aligned} S &= \int_{-2}^2 (|4t - t^3|2t) dt \\ &= \int_{-2}^2 (8t^2 - 2t^4) dt \\ &= \left[\frac{8t^3}{3} - \frac{2t^5}{5} \right]_{-2}^2 \\ &= \left(\frac{64}{3} - \frac{64}{5} \right) - \left(\frac{-64}{3} + \frac{64}{5} \right) \\ &= 128 \left(\frac{5}{15} - \frac{3}{15} \right) \\ &= \frac{256}{15} \end{aligned}$$

References

5E-1-luko.de.vriendt.INF.jpg